Cambridge (CIE) A Level Maths: Probability & Statistics 1

Your notes

Binomial & Geometric Distribution

Contents

- The Binomial Distribution
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Properties of Binomial Distribution

What is a binomial distribution?

- A binomial distribution is a **discrete probability distribution**
- The **discrete random variable *X*** follows a **binomial distribution** if it **counts the number of successes** when an experiment satisfies the conditions:
  - There are a **fixed finite number of trials** (n)
  - The outcome of each trial is **independent** of the outcomes of the other trials
  - There are **exactly two outcomes** of each trial (**success or failure**)
  - The **probability of success (p) is constant**
- If *X* follows a binomial distribution then it is denoted $X \sim B(n,p)$
  - n is the number of trials
  - p is the probability of success
- The **probability of failure is 1−*p*** which is sometimes denoted as ***q***
- The formula for the probability of *r* successful trials is given by:
  - $P = (X = r) = \binom{n}{r} p^r (1-p)^{n-r}$ for $r = 0, 1, 2, \ldots, n$
  - This is equal to the term which includes p^r in the expansion of $(p+q)^n$ where $q = 1 - p$ (this shows the link with the Binomial Expansion)

What are the important properties of a binomial distribution?

- The **expected number (mean)** of successful trials is np
- The **variance** of the number of successful trials is $np(1-p)$
  - Square root to get the standard deviation
- The distribution can be represented visually using a vertical line graph
  - If *p* is **close to 0** then the graph has a **tail to the right**
  - If *p* is **close to 1** then the graph has a **tail to the left**
  - If *p* is **close to 0.5** then the graph is **roughly symmetrical**
  - **If *p* = 0.5** then the graph is **symmetrical**

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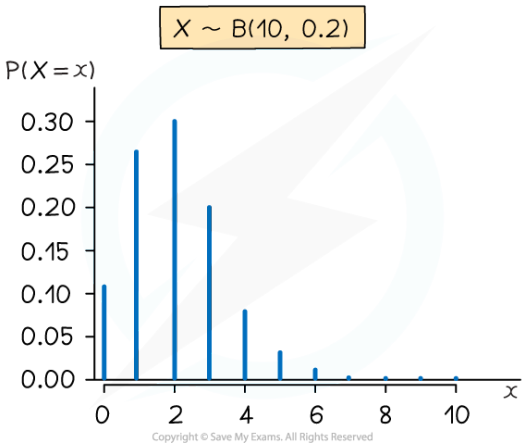

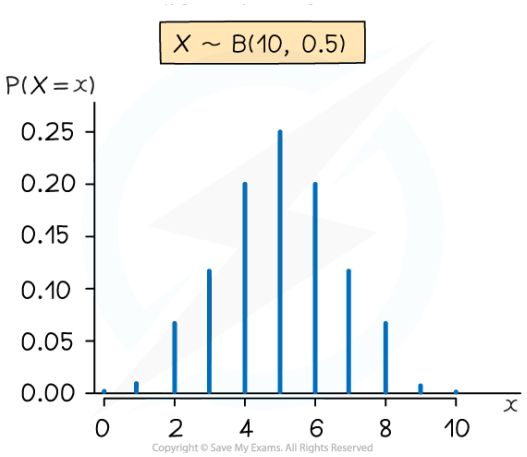

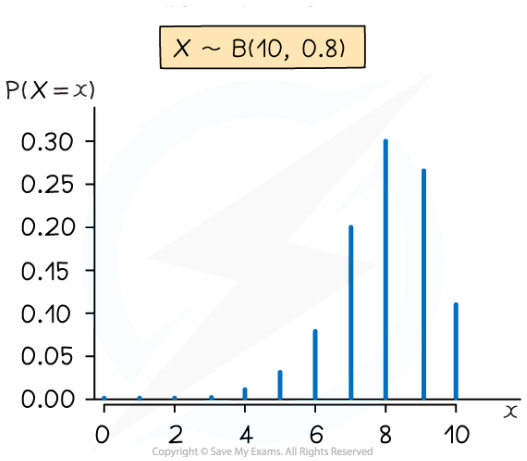

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Modelling with Binomial Distribution

How do I set up a binomial model?

- **Identify** what a **trial** is in the scenario

- For example: rolling a dice, flipping a coin, checking hair colour
- **Identify** what the **successful outcome** is in the scenario
  - For example: rolling a 6, landing on tails, having black hair
- Make sure you **clearly state** what your **random variable** is
  - For example, let X be the number of students in a class of 30 with black hair

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What can be modelled using a binomial distribution?

- Anything that satisfies the **four conditions**
- For example, let T be the number of times a fair coin lands on tails when flipped 20 times: $T \sim \mathrm{B}\left(20, \frac{1}{2}\right)$
  - A trial is flipping a coin: There are 20 trials so $n = 20$
  - We can assume each coin flip does not affect subsequent coin flips: They are independent
  - A success is when the coin lands on tails: Two outcomes - tails or not tails (heads)
  - The coin is fair: The probability of tails is constant with $p = \frac{1}{2}$
- Sometimes it might seem like there are more than two outcomes
  - For example, let Y be the number of yellow cars that are in a car park full of 100 cars
  - Although there are more than two possible colours of cars, here the trial is whether a car is yellow so there are two outcomes (yellow or not yellow)
  - Y would still need to fulfil the other conditions in order to follow a binomial distribution
- Sometimes a sample may be taken from a population
  - For example, 30% of people in a city have blue eyes, a sample of 30 people from the city is taken and X is the number of them with blue eyes
  - As long as the population is large and the sample is random then it can be assumed that each person has a 30% chance of having blue eyes

What can not be modelled using a binomial distribution?

- Anything where the number of trials is **not fixed** or is **infinite**
  - The number of emails received in an hour
  - The number of times a coin is flipped until it lands on heads
- Anything where the outcome of **one trial affects** the outcome of the **other trials**
  - The number of caramels that a person eats when they eat 5 sweets from a bag containing 6 caramels and 4 marshmallows

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- If you eat a caramel for your first sweet then there are less caramels left in the bag when you choose your second sweet

- Anything where there are **more than two outcomes** of a trial
  - A person's shoe size
  - The number a dice lands on when rolled
- Anything where the **probability of success changes**
  - The number of times that a person can swim a length of a swimming pool in under a minute when swimming 50 lengths
    - The probability of swimming a lap in under a minute will decrease as the person gets tired

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Worked Example

It is known that 8% of a large population are immune to a particular virus. Mark takes a sample of 50 people from this population. Mark uses a binomial model for the number of people in his sample that are immune to the virus

(a) State the distribution that Mark uses.

(b) State the two assumptions that Mark must make in order to use a binomial model.

a) A trial is checking if a person is immune to the virus
A success is if the person is immune.
Let X be the number of people in the sample immune to the virus

$X \sim B(50, 0.08)$

Number of people in sample

Probability of being immune to the virus

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b) Mark needs to assume that:
- each person in the population has an 8% chance of being immune
- the sample is random and the people are independent a person being immune does not affect the immunity of others

For example:
if all 50 came from the same family then they would not be independent

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Examiner Tips and Tricks

- If you are asked to criticise a binomial model always consider whether the trials are independent, this is usually the one that stops a variable from following a binomial distribution!

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Calculating Binomial Probabilities

Throughout this section we will use the random variable $X \sim \mathrm{B}(n,p)$. For binomial, the probability of a *X* taking a non-integer or negative value is always zero. Therefore any values mentioned in this section will be assumed to be non-negative integers.

Where does the formula for a binomial distribution come from?

- The formula for calculating an individual **binomial probability** is
  - $\mathrm{P}(X=r)=p_r=\binom{n}{r}p^r(1-p)^{n-r}$
    - If there are *r* successes then there are $(n-r)$ failures
    - The number of times this can happen is calculated by the **binomial coefficient**
    - $\binom{n}{r}={}^nC_r=\frac{n!}{r!(n-r)!}$
- This can be seen by considering a **probability tree diagram** with *n* trials, where *p* is the probability of success and the tree diagram is being used to find *r* successes
- $\binom{n}{r}$ is the number of pathways through the tree there would be exactly *r* successes within the *n* trials
- The formula allows statisticians to quickly find probabilities for larger values of *n* without needing to draw the whole tree diagram
- Your calculator may have a function that would allow you to calculate binomial probabilities
  - You can learn how to use this to check your work but it is important you always show your working using the formula to get the marks in the exam

How do I calculate the cumulative probabilities for a binomial distribution?

- Most of the time you will be required to calculate **cumulative binomial probabilities** rather than individual ones
- Use the formula to find the individual probabilities and then add them up
- Make sure you are confident working with inequalities for **discrete values**
  - Only integer values will be included so it is easiest to look at which integer values you should include within your calculation

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- Sometimes it is quicker to find the probabilities that are **not being asked for** and subtract from one
- $P(X \leq r)$ is asking you to find the probabilities of **all values up to and including *r***
  - This means all values that are **at most *r***
  - Don't forget to include P(*X* = 0)
  - It could also be written as $P(X \leq r) = 1 - P(X > r)$
- $P(X < r)$ is asking you to find the probabilities of **all values up to but not including *r***
  - This means all values that are **less than *r***
  - Stop at *r* - *1*
  - It could also be written as $P(X < r) = 1 - P(X \geq r)$
- $P(X \geq r)$ is asking you to find the probabilities of **all values greater than and including *r***
  - This means all values that are **at least *r***
  - It could also be written as $P(X \geq r) = 1 - P(X < r)$
- $P(X > r)$ is asking you to find the probabilities of **all values greater than but not including *r***
  - This means all values that are **more than r**
  - Start at *r* + *1*
  - It could also be written as $P(X > r) = 1 - P(X \leq r)$
- If calculating $P(a \leq X \leq b)$ pay attention to whether the probability of *a* and *b* should be included in the calculation or not
  - For example, $P(4 < X \leq 10)$:
    - You want the integers 5 to 10

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Worked Example

If X is the random variable $X \sim B(10, 0.35)$. Find:

(i) $P(X = 3)$

(ii) $P(X \leq 3)$

(iii) $P(X > 3)$

(iv) $P(3 < X < 6)$

$X \sim B(10, 0.35)$

$n = 10$ $\quad$ $p = 0.35$

(i) $P(X = 3)$

$x = 3$

Use the nC_r button on your calculator

$$P(X = 3) = \binom{10}{3}(0.35)^3(1 - 0.35)^{10-3}$$

$$= (120)(0.35)^3(0.65)^7$$

$$= 0.2522...$$

$$P(X = 3) = 0.252 \text{ (3sf)}$$

(ii) $P(X \leqslant 3)$ This means find the probability X is at most 3, find the probability $x = 0, 1, 2$ or 3

$$P(X \leqslant 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

$$= \binom{10}{0}(0.35)^0(0.65)^{10} + \binom{10}{1}(0.35)^1(0.65)^9$$

These are both 1

$$+ \binom{10}{2}(0.35)^2(0.65)^8 + \binom{10}{3}(0.35)^3(0.65)^7$$

$$= 0.65^{10} + (10)(0.35)(0.65)^9 + (45)(0.35)^2(0.65)^8 + 0.2522...$$

$$= 0.513827...$$

Type everything into your calculator at once and check it before pressing =

$$P(X \leqslant 3) = 0.514 \quad \text{(3sf)}$$

(iii) $P(X > 3)$ This includes all integers greater than 3, this would take a lot of calculations so instead use $P(X > x) = 1 - P(X \leqslant x)$

$$P(X > 3) = 1 - P(X \leqslant 3)$$

$$= 1 - 0.513827... \qquad = 0.486172...$$

$$P(X > 3) = 0.486 \quad \text{(3sf)}$$

(iv) $P(3 < X < 6)$ This is all the integers between 3 and 6 but not including 3 and 6

$$P(3 < X < 6) = P(X = 4) + P(X = 5)$$

$$= \binom{10}{4}(0.35)^4(0.65)^6 + \binom{10}{5}(0.35)^5(0.65)^5$$

$$= 0.391238...$$

$$P(3 < X < 6) = 0.391 \quad \text{(3sf)}$$

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Examiner Tips and Tricks

- Looking carefully at the inequality within the probability is key here, make sure you consider which integers should be counted within your calculations.

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Properties of Geometric Distribution

What is a geometric distribution and its notation?

- A **geometric distribution** is a **discrete probability distribution**
- The discrete random variable X follows a geometric distribution if it counts the number of trials until the first success occurs for an experiment that satisfies the conditions
  - **Each trial has only two outcomes** Broadly labelled as "success" and "failure" - these are mutually exclusive

Such trials may be referred to as **Bernoulli trials** – named after the Swiss mathematician Jacob Bernoulli (1655–1705)

- **The outcomes of trials are independent** The outcome of one trial does not affect the outcome of another trial
  - **The probability of each outcome is constant across all trials**

i.e. The probability of "success" does not change between trials

- *p* is the probability of "**success**" in a **single trial**
  - The probability of "**failure"** in a **single** trial is *1– p* ; often denoted by *q*
- If *X* follows a geometric distribution, it is denoted by $X \sim \text{Geo}(p)$
- The formula for finding the probability that the first success occurs after *r* trials (or after *r* -1 failures) is

$$P(X = r) = (1 - p)^{r-1}p$$

- e.g. If the probability of success in a single trial is 0.3, then the probability it will take 6 trials to obtain the first success is given by

$$P(X = 6) = (1 - 0.3)^{6-1}(0.3) = (0.7)^5(0.3) = 0.050421$$

What does a geometric distribution look like?

- If represented visually, using a vertical line graph, the probabilities in a geometric distribution decrease but **never** reach zero
  - The probabilities form a **geometric progression**
    - Question: What would the first term ("a"), the common ratio ("r") and the sum to infinity be ("S_∞")? (Answer below diagram)
    - The probabilities decrease exponentially; drawing a curve through the tops of the lines would produce a decreasing exponential curve
  - The graphs also show how 1 is the **mode** for **every** geometric distribution

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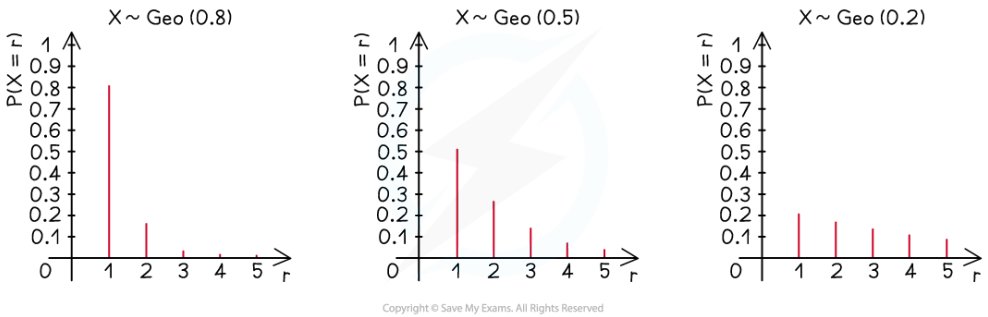

- Answer: $a = p, r = q = 1 - p,\ S_{\infty} = 1$

What are the properties of a geometric distribution?

- $\mathrm{P}(X = r) > 0$ for all r
  - Every geometric distribution has an infinite (discrete) **sample space** which is the set of natural numbers ($\mathbb{N}$) or positive integers ($\mathbb{Z}^+$)
- $\mathrm{P}(X = r) < \mathrm{P}(X = r - 1)$ for all r
  - The **mode** of **every** geometric distribution is **1** (the value of r that has the highest probability)
- Geometric distributions are **memoryless**
  - The number of trials needed for the first success is not dependent on the number of trials that have already occurred e.g. If 5 (failed) trials have already occurred, the probability of the first success happening after 7 trials is the same as the probability of it happening after 2 trials which would be $(1 - p)p$
  - Mathematically this is written as

$\mathrm{P}(X = r | X > k) = \mathrm{P}(X = r - k)$ where k is the number of trials that have already occurred

e.g. $\mathbf{P(X = 7 | X > 5)} = \mathrm{P}(X = 7 - 5) = \boldsymbol{P(X = 2)}$

- Geometric distributions have the **recurrence relation** $P(X = r) = qP(X = r - 1)$
- The **mean**, or **expected value**, of a geometric distribution is $\boldsymbol{E(X) = \frac{1}{p}}$

Modelling with Geometric Distribution

How do I set up a geometric model?

- **Identify** what a **trial** is in the **context** of a problem
  - Flipping a coin, rolling a dice, a football match
- **Identify** a **successful outcome**
  - Heads, a square number, a win

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- **Identify** the probability of "**success**"
  - $0.5, \frac{1}{3}, 42\%$
- Define your **random variable** using the correct **notation**
  - Let X be the number of trials required to obtain the first heads when flipping a fair coin, $X \sim \text{Geo}(0.5)$

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What can be modelled using a geometric distribution?

- Anything where the **first** occurrence of a **successful outcome** is of significance
  - Rolling a double with two dice before being allowed to start a game
  - The number of on/off presses a switch can withstand before wearing out

(In which case the first "success" would be the first failure of the switch!)

- In addition, the scenario must satisfy the t**hree conditions**
  - Trials only have two outcomes of interest
  - Trials are independent
  - Probability of success is constant for all trials
  - These are also three of the four conditions for a **binomial distribution** so are not enough on their own – it will also depend on the context
- Many scenarios may appear as having more than two outcomes but in the context of the question only two are of significance
  - e.g. A light that randomly flashes in 8 different colours, but the only colour of interest is blue So "blue" is "success" and all other colours, regardless of whether it is red, yellow, etc – i.e. "not blue" - is "failure"
- Sometimes a sample may be taken from a population
  - As long as the population is large enough and the sample is random the probability of "success" in the sample is the same as the probability of "success" in the population

What cannot be modelled using a geometric distribution?

- Be careful not to confuse **binomial** and **geometric** distributions/models
  - **Binomial** is for the number of **successes** in a **fixed** number of **trials**
  - **Geometric** is for the number of **trials** up to and **including** the **first success**
- Anything where a trial would have **more than two outcomes** of interest
  - e.g. Outcome of a football match – win, draw or lose
- Where the **probability** of an **outcome** of a **trial** is **influenced** by a previous trial
  - i.e. trials are **not independent**

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- e.g. drawing counters from a bag **without** replacement
- Anything where the **probability** of "**success**" **changes** with time – or practice
  - e.g. a skateboarder performing a trick - the probability of success should increase after practising the trick

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Calculating Geometric Probabilities

How do I calculate geometric probabilities?

- Identify *p*, the probability of "success" and *1– p*, the probability of "failure" (q)
- For **exact** probabilities use $\mathrm{P}(X=r)=(1-p)^{r-1}p$
- For **inequalities** use
  - $\mathrm{P}(X>r)=(1-p)^{r}$
    - This means the first success occurs ***after r*** trials, therefore the first r trials all ended in failure
    - Similarly, $\mathrm{P}(X\geq r)=\mathrm{P}(X>r-1)=q^{r-1}$
  - $\mathrm{P}(X<r)=1-\mathrm{P}(X>r-1)=1-q^{r-1}$
    - Similarly, $\mathrm{P}(X\leq r)=1-\mathrm{P}(X>r)=1-q^{r}$
  - $\mathrm{P}(a\leq X\leq b)=\mathrm{P}(X\leq b)-\mathrm{P}(X<a)$
    - If *a* and *b* are close it may be easier to use

$$P(X=a)+P(X=a+1)+\ldots+P(X=b)$$

- Logic can be used to deduce most geometric distribution questions so memorising these formulae is not essential
- Beware of questions that exploit the ***memorylessness*** property of geometric distributions – loosely called "given that" questions
  - e.g. $P(X=8|X>6)$ means "the probability that equals 8 given that is greater than 6" or "the probability of 8 trials given that 6 trials have already occurred"

$$P(X=8|X>6)=P(X=8-6)=P(X=2)$$

- The **mean** (**expected value**) or **mode** of a geometric distribution may be required
  - $\mathrm{E}(X)=\dfrac{1}{p}$
  - The **mode** is **1** for all geometric distributions

Worked Example

Given that $X\sim\mathrm{Geo}(0.4)$ find

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(i) $P(X=5)$

(ii) $P(X>5)$

(iii) $P(3 \le X < 8)$

(iv) $P(X=8 \mid X>5)$

(v) The mode of X.

Identify p and q

$p = 0.4 \qquad \therefore q = 1 - 0.4 = 0.6$

(i) $P(X = 5) = (0.6)^4(0.4) = 0.05184$ $\qquad$ $P(X = 5) = 0.052$ (3 s.f.)

(ii) $P(X > 5) = (0.6)^5 = 0.07776$ $\qquad$ $P(X > 5) = 0.078$ (3 s.f.)

There will be 5 "failures"

(iii) $P(3 \leqslant X < 8) = P(X < 8) - P(X < 3)$

$= 1 - P(X > 7) - [1 - P(X > 2)]$

$= P(X > 2) - P(X > 7)$

$= (0.6)^2 - (0.6)^7 = 0.332006...$

$P(3 \leqslant X < 8) = 0.332$ (3 s.f.)

(iv) $P(X = 8 \mid X > 5) = P(X = 8 - 5) = P(X = 3)$

Geometric distribution is memoryless

"Given that"

$= (0.6)^2(0.4)$

$= 0.144$

$P(X = 8 \mid X > 5) = 0.144$

(v) The mode of **all** geometric distributions is 1

Mode of X is 1

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Examiner Tips and Tricks

- Try not to get bogged down with formulae for the geometric distribution, most questions can be deduced using logic
- If you are asked to criticise a geometric model always consider whether trials are independent
  - especially if it involves "practising" or "performing" a skill
  - most people will improve after they've made several attempts at a skill
  - so the probability of success should gradually increase over time
- If finding the number of trials required (r) then be careful counting calculator presses; remember you are likely to be finding $r - 1$ (the number of failures

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before success) in the first instance

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